

Report of Project 2: Chemical Kinetics

Russ, Rachel

*Department of Aeronautics and Astronautics
Purdue University
February 19, 2026*

1 Stiff ODEs

1.1 Non-Stiff Differential Functions – Q1(c)

There are two ways to formulate the reaction progress variable. This variable is used to track the progress of the reaction from reactants to products. A value of 0 means the mixture is fully unreacted while a value of 1 means the reaction is fully reacted. These equations are shown in Eq.(1) and Eq.(2).

$$\frac{dc}{dt} = S(c) = \omega_c A c^n (1 - c) \quad (1)$$

$$\frac{dc}{dt} = \omega_c r \bar{A} c^p (1 - c)^{\frac{1}{2}} \quad (2)$$

Where for Eq.(2),

$$\frac{dr}{dt} = \omega_r (R(c) - r) \quad (3)$$

$$R(c) \equiv B c^q (1 - c)^{\frac{1}{2}} \quad (4)$$

The variable A can be found by $\max\{S(c)\} = \omega_c$. B can be found by $\max\{R(c)\} = 1$, and $\bar{A}$ can be found by $r = R(c)$, $\max\{dc/dt\} = \omega_c$. It can be assumed that p=4 and q=2. The variable r(t), found by solving the differential equation Eq.(3), is the normalized radical mass fraction. R(c), the value found by Eq.(4), is the partial equilibrium value. The normalized mass fraction relaxes towards the steady state (partial equilibrium value) at a rate of ω_r where ω_r is an inverse time scale, as can be used to adjust stiffness.

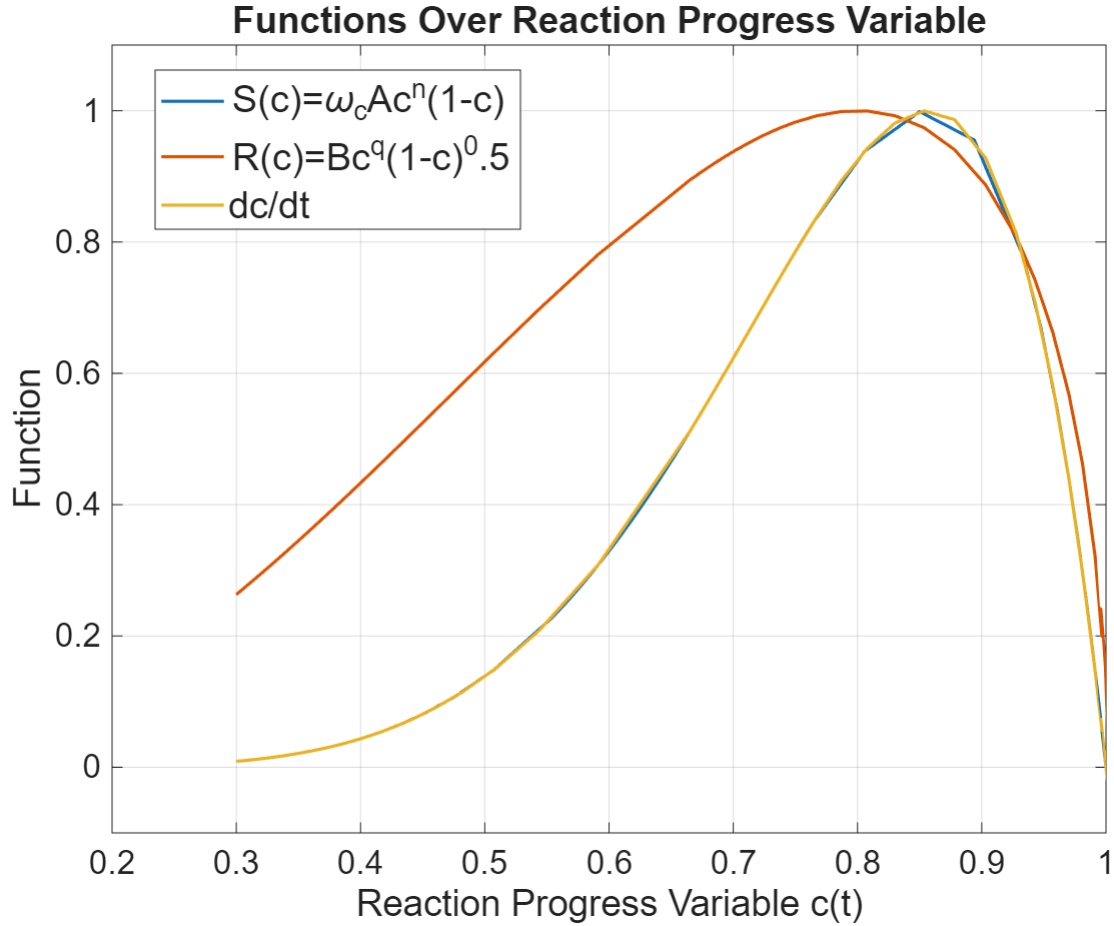


Figure 1 Differential Functions as a Function of Reaction Progress Variable

Fig. 1 shows $S(c)$, $R(c)$, and the change in the reaction progress variable with respect to time (dc/dt) plotted over the reaction progress variable. All three of these models have a maximum value of 1 at c values between 0.8 and 0.9 meaning the mixture is almost fully reacted. It can be noted that dc/dt and $S(c)$ are approximately the same for each reaction progress variable. This is since as $\omega_r \rightarrow \infty$, Eq.(3) becomes $r = R(c)$, and Eq.(2) becomes Eq.(5).

$$\frac{dc}{dt} = \omega_c R(c) \bar{A} c^p (1-c)^{\frac{1}{2}} = \omega_c A c^{p+q} (1-c) \quad (5)$$

Eq.(5) is identical to Eq.(1) if $p+q=n$. The data for the graph satisfies this equation which makes $S(c)$ and dc/dt the same.

1.2 Non-Stiff Equation Solver with ($\omega_r = 10$) - Q1(d)

Eq.(2) and Eq.(3) can be solved using a differential equation solver with different ω_r values. Fig. 2 shows these two equations solved with $\omega_r = 10$ and with the non-stiff differential equation solver ODE 45 in MATLAB.

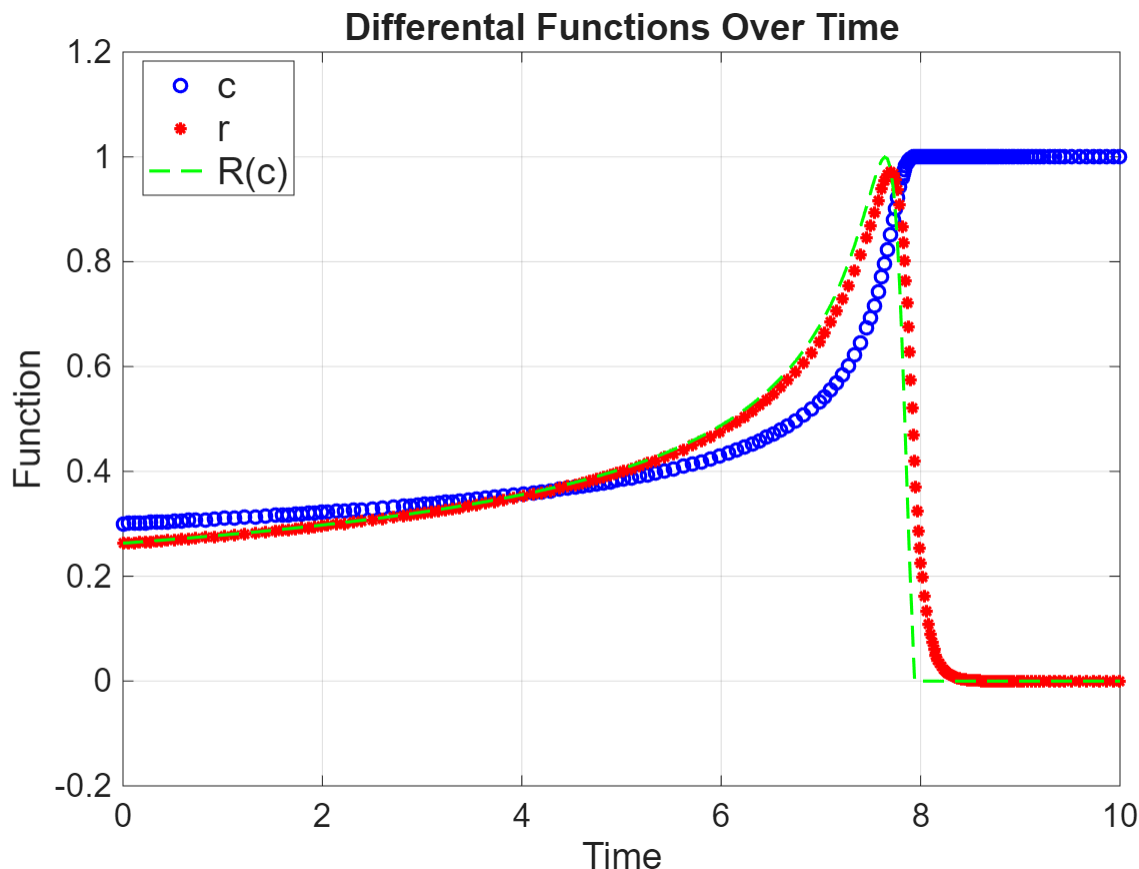


Figure 2 Differential Functions over Time (Relatively Non-Stiff)

This value of ω_r is relatively non-stiff. This means that the non-stiff equation solver is adequate for the evaluation of these equations. Fig. 2 shows that the reaction is finished at a time of 8 seconds. This is because this is the time that the c function has a value of 1, representing that all the reactants have turned into products. The initial condition for $r(0)$ is 0.2631. This is found with inputs of $c(0)$ and q where $c(0)$ is 0.3 and q is 2.

Table 1 Solver Statistics for a Non-Stiff Solver ($\omega_r = 10$)

Elapsed Time (sec)	Total Number of Steps	Successful Steps	Failed Steps	Function Evaluations
0.005674	55	47	8	331

Table 1 shows the statistics for the non-stiff ODE solver. There is a relatively low number of function evaluations showing that the cost to run this program is lower. The program also runs in an acceptable amount of time. There are enough successful steps that the program can be considered successful. From this information, it can be concluded that with this ω_r value, a non-stiff ODE solver can be used.

1.3 Non-Stiff Equation Solver with ($\omega_r = 10^3$) - Q1(e)

Eq.(2) and Eq.(3) can be solved using a differential equation solver with different ω_r values. Fig. 3 shows these two equations solved with $\omega_r = 1000$ with the non-stiff differential equation solver ODE 45 in MATLAB.

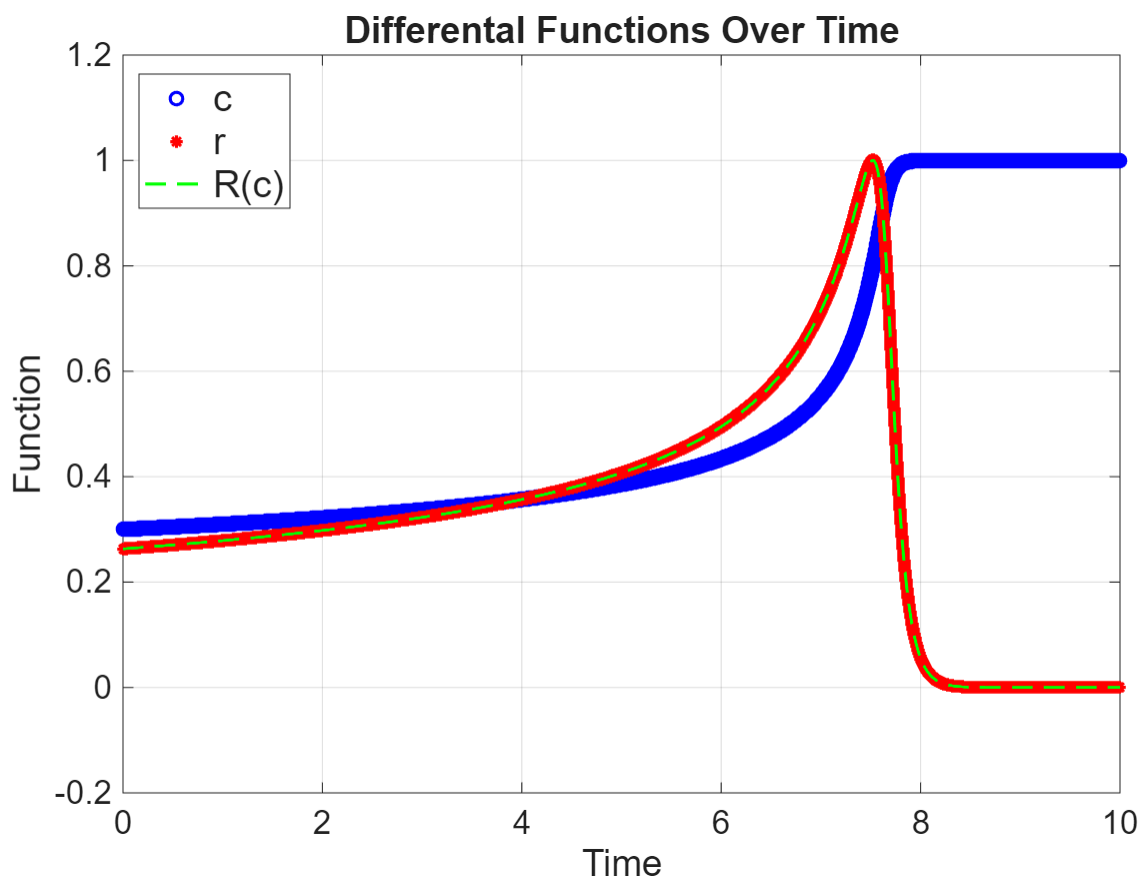


Figure 3 Differential Functions over Time (Relatively Stiff)

This value of ω_r is relatively stiff. This means that the non-stiff equation solver is not as adequate for the evaluation of these equations as it was for the lower ω_r that was used in section 1.2. Fig. 3 shows that the reaction is finished at a time of 7.5 seconds. This is because this is the time that the c function has a value of 1, representing that all the reactants have turned into products. The initial condition for $r(0)$ has not changed from the previous subpart (1.2).

Table 2 Solver Statistics for a Non-Stiff Solver ($\omega_r = 1000$)

Elapsed Time (sec)	Total Number of Steps	Successful Steps	Failed Steps	Function Evaluations
0.069010	3196	3017	179	19177

Table 2 shows the statistics for the non-stiff ODE solver. There is a much higher number of function evaluations compared to $\omega_r = 10$ evaluation, in fact, there is a 193% of the number of evaluations. There are also a much larger number of total steps with the same percent increase of

193%. This program also takes more time to run compared to the lower ω_r value. However, there are still enough successful steps compared to failed steps to consider this solver successful. These major increases in the number of evaluations and number of steps show that this model is much more expensive to run compared to the less-stiff model. From this information, it can be concluded that with this ω_r value, a non-stiff ODE solver should not be used.

1.4 Stiff Equation Solver with ($\omega_r = 10^3$) - Q1(f)

Eq.(2) and Eq.(3) can be solved using a differential equation solver with different ω_r values. Fig. 4 shows these two equations solved with $\omega_r = 1000$ with the stiff differential equation solver ODE 23s in MATLAB.

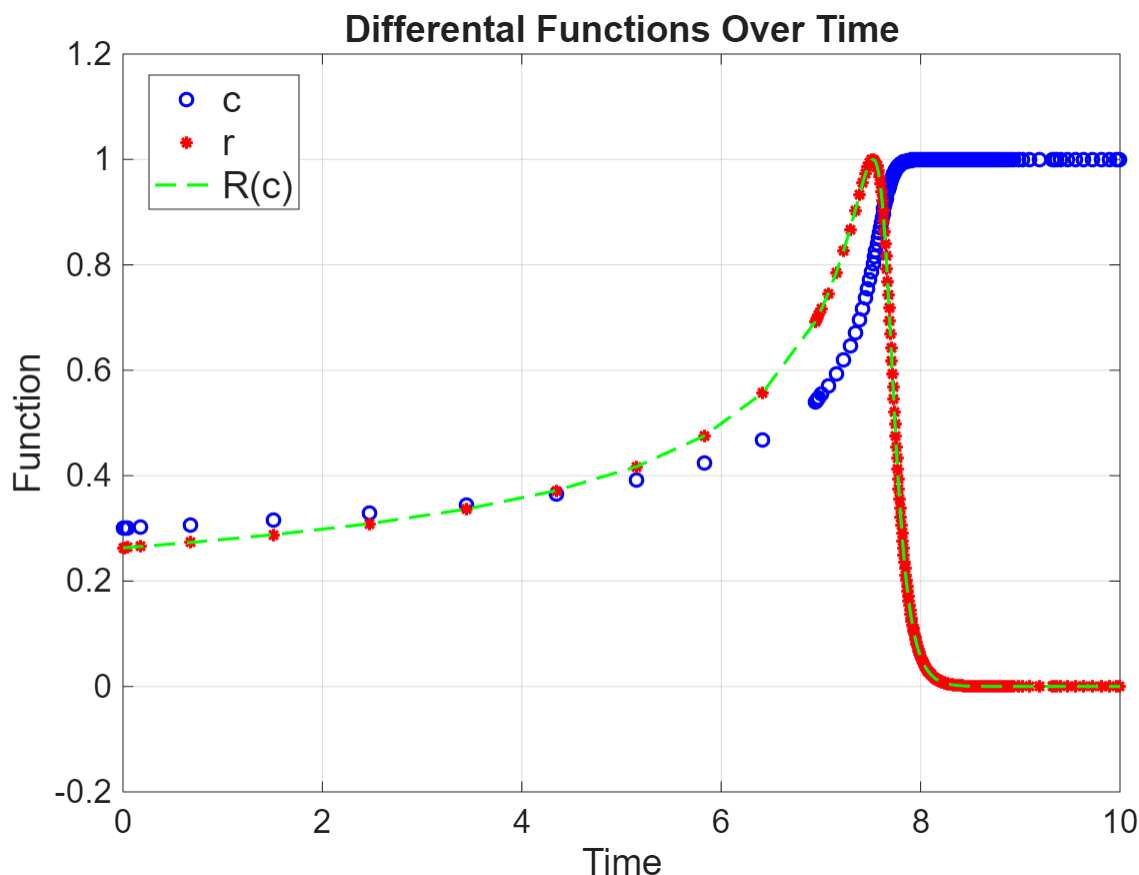


Figure 4 Differential Functions over Time (Relatively Stiff)

This value of ω_r is relatively stiff. This means that the non-stiff equation solver is not as adequate for the evaluation of these equations; however, the stiff equation solver works quite well as shown by Fig. 4. The figure shows that the reaction is finished at a time of 7.5 seconds which matches the value that the non-stiff equation software got for the same ω_r value. This is because this is the time that the c function has a value of 1, representing that all the reactants have turned into products. The initial condition for r(0) has not changed from the previous subparts (1.2 and 1.3).

Table 3 Solver Statistics for a Stiff Solver ($\omega_r = 1000$)

Elapsed Time (sec)	Total Number of Steps	Successful Steps	Failed Steps	Function Evaluations	partial derivatives	LU decompositions	solutions of linear systems
0.327089	238	224	14	1150	224	238	714

Table 3 shows the statistics for the stiff ODE solver. These values can be compared to the non-stiff ODE solver to show that the stiff ODE solver is less costly and the better option for more stiff equations. The total number of steps has significantly decreased between the solvers with a 172% decrease from the non-stiff solver to the stiff solver. The number of function evaluations also decreased by 177% between the two solvers. This shows that stiff solver is the least costly option between the two solvers. However, interesting enough, the elapsed time for the stiff solver is higher compared to the non-stiff solver. This is since the system is not that stiff, so the non-stiff solver can still handle the task.

1.5 Comparison of Stiff and Non-Stiff ODE Solvers ($\omega_r = 10^6$) - Q1(g)

Eq.(2) and Eq.(3) can be solved using a differential equation solver with different ω_r values. Fig. 5 shows these two equations solved with $\omega_r = 10^6$ with the stiff differential equation solver ODE 23s and the non-stiff differential equation solver ODE 45 in MATLAB.

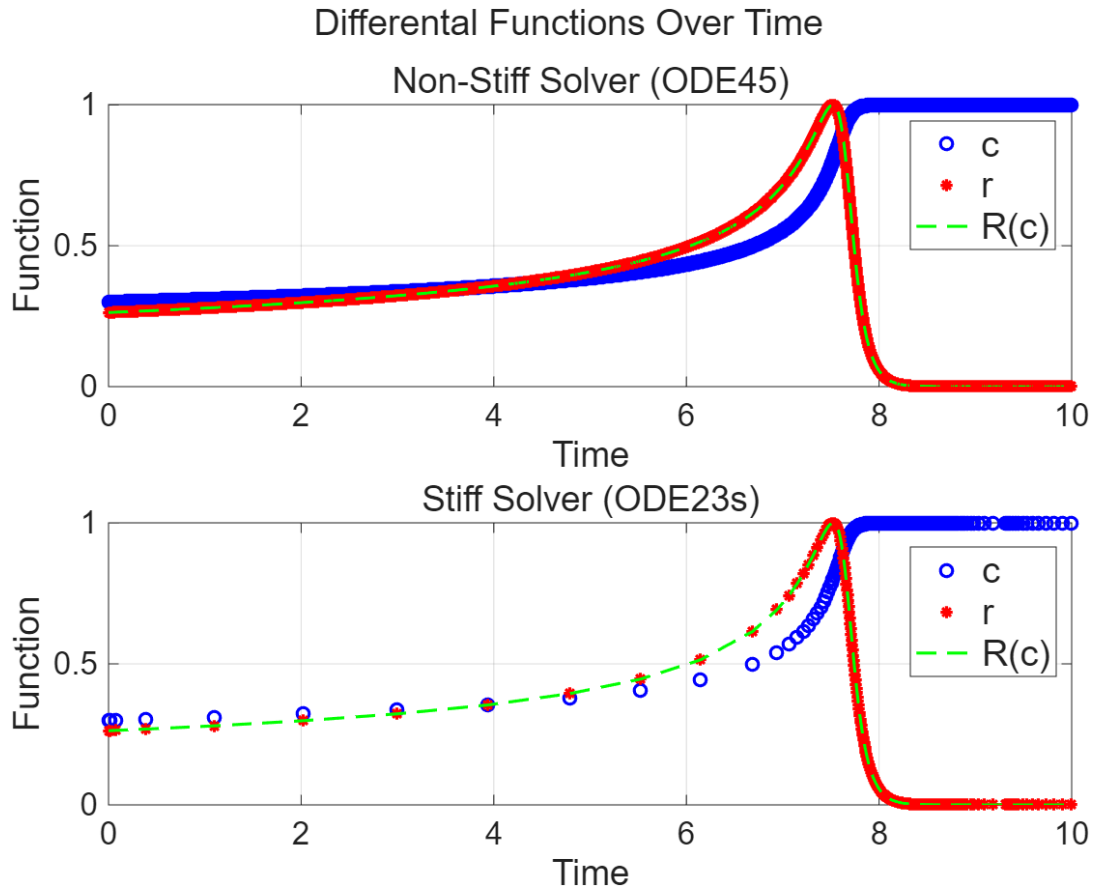


Figure 5 Comparison of Non-Stiff and Stiff ODE solver for Stiff System

This value of ω_r is stiff. This means that the non-stiff equation solver is not as adequate for the evaluation of these equations; however, the stiff equation solver works quite well as shown by Fig. 5. The figure shows that the reaction is finished at a time of 7.5 seconds which matches the value that the non-stiff equation software got for the same ω_r value. This is because this is the time that the c function has a value of 1, representing that all the reactants have turned into products. The initial condition for $r(0)$ has not changed from the previous subparts (1.2 and 1.3 and 1.4). Both solvers show the same data with the values being the same of each. This shows that both solvers are technically correct in their evaluation.

Table 4 Solver Statistics for a Stiff Solver ($\omega_r = 1000000$)

Solver	Elapsed Time (sec)	Total Number of Steps	Successful Steps	Failed Steps	Function Evaluations
Stiff (ODE 23s)	0.022496	305	282	23	1,458
Non-Stiff (ODE 45)	67.786282	3,214,008	3,013,170	200,838	19,284,000

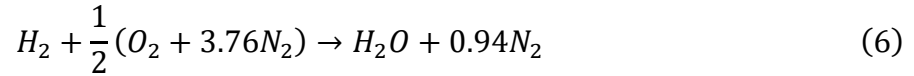
Table 4 shows the statistics for the stiff and non-stiff ODE solver. The total number of steps has significantly different results between the solvers with a 200% decrease from the non-stiff solver

to the stiff solver. The number of function evaluations also decreased by 200% among the two solvers. This shows that the stiff solver is by far the least costly option between the two solvers. The elapsed time is also significantly different between the two solvers with also a 200% difference. It can be concluded that the stiffer the system is, the more important it is to use a stiff solver to decrease time and cost of the evaluation even though the final results are the same.

2 H₂ – Autoignition

2.1 Reactive Species Mass Fractions as Functions of Time – Q2(a)

The autoignition of hydrogen and air can be represented by the chemical equation in Eq.(6) below where it is assumed that there is one mole of fuel.



The mass of each species (m_α) can be calculated in Eq.(7) where MW_α is the molecular weight of that species and N_α is the molar amounts of each species in the stoichiometric equation.

$$m_\alpha = MW_\alpha N_\alpha \quad (7)$$

Eq.(8) shows the equation for mass fraction of each species (Y_α) where m is the total weight of all the species (70.64 g).

$$Y_\alpha = \frac{m_\alpha}{m} \quad (8)$$

Table 5 shows these mass fractions calculated for each of the autoignition species that were put into FLUENT for initiation.

Table 5 Mass Fractions of Species Before Reaction

Mass Fraction of H ₂	Mass Fraction of O ₂	Mass Fraction N ₂
0.0283	0.2265	0.745

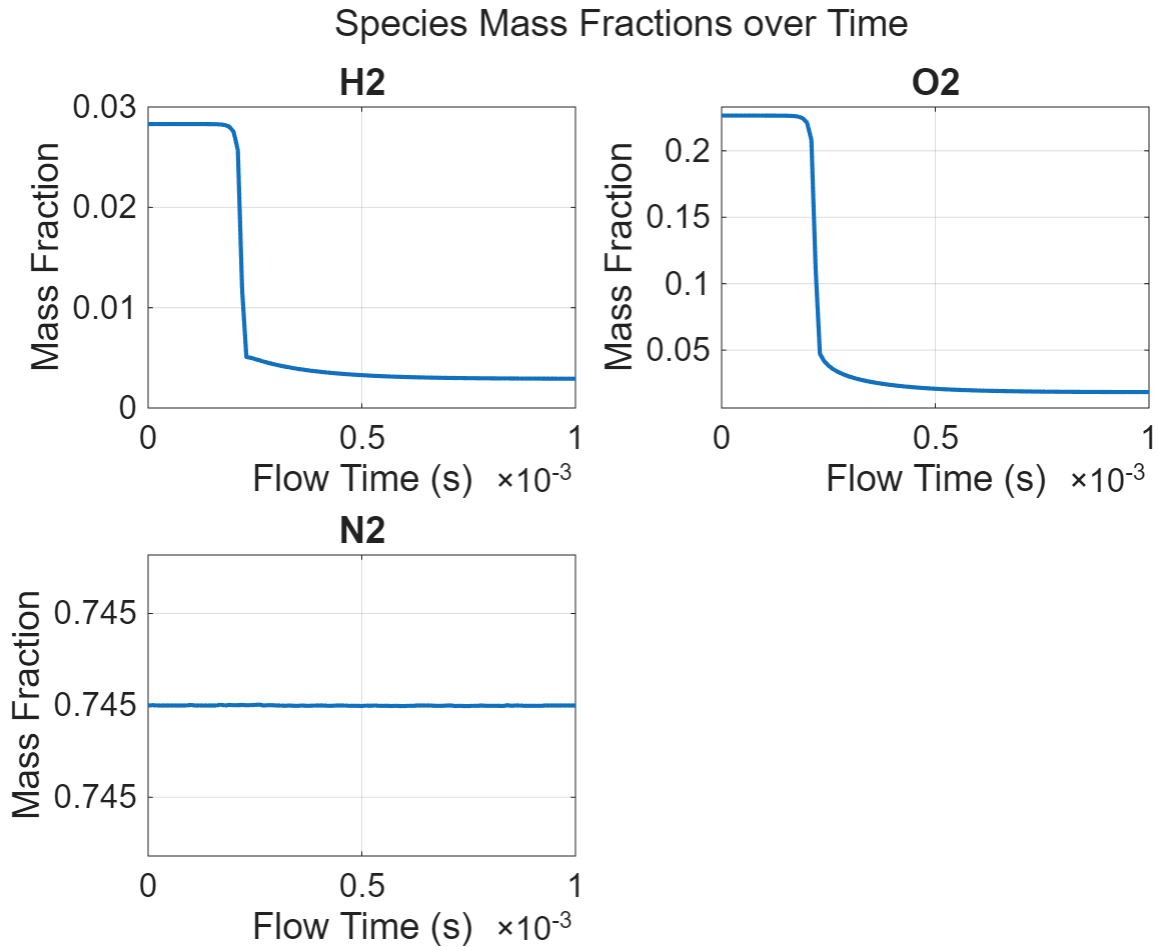


Figure 6 Species Mass Fractions (H_2 , O_2 , and N_2) Over Flow Time

As the reaction progresses, H_2 and O_2 get used in creating the products. The major drop of the mass fraction is at a flow time of around 0.00025 seconds. One thing to note is that the mass fractions for H_2 and O_2 are not zero once the reaction is complete. This is because not all these species are used to create H_2O as the reaction is not perfect. N_2 stays constant in this process because there is N_2 in the products only. This is because all the N_2 in the reactants goes into the N_2 in the products.

2.2 Numerical Accuracy of Results – Q2(b)

Qualitatively, the data makes sense and seems correct; however, it is hard to tell if it actually is quantitatively because there is nothing for it to be based on. The numerical accuracy of the results heavily depends on the step size as this means there will be a different number of equations solved. The equations being solved in the first place are Navier-Stokes equations for momentum, continuity, and enthalpy. It also considers species mass fraction transport and reaction rates. A smaller step size means more accurate results. However, this means the processing costs are more as there are more equations being solved.

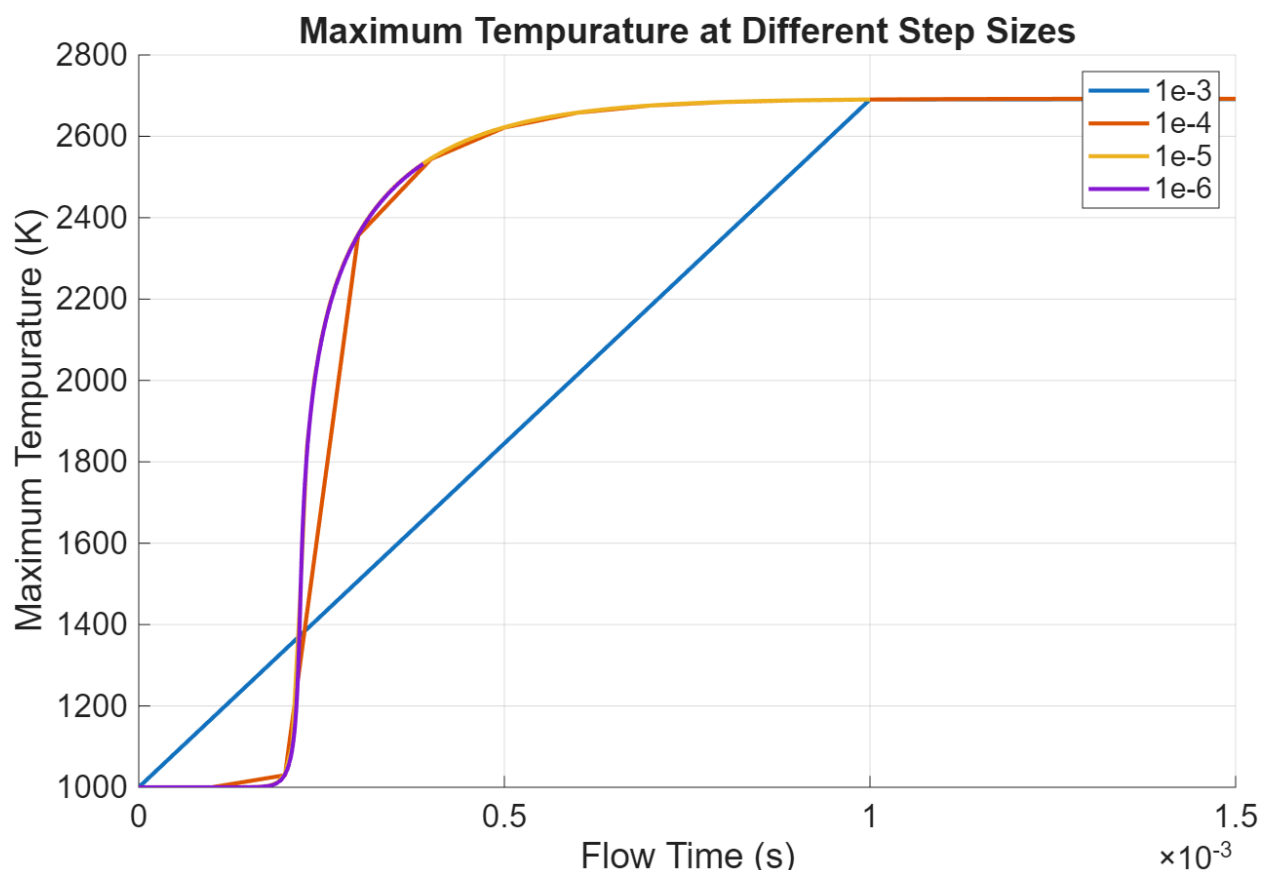


Figure 7 Maximum Temperature of Reaction for Different Step Sizes

Fig. 7 shows the maximum temperature of the reaction with differing step sizes. Clearly, the smaller the step size the more accurate the results as the results are smoother. Table 6 shows the percent differences between each of the step sizes at which time in the flow a maximum temperature of 1800 K occurs. This temperature was chosen because it is the temperature in the middle of the reaction.

Table 6 Percent Differences in Flow Time at a Maximum Temperature of 1800K

1e-3 $\rightarrow$ 1e-4	1e-4 $\rightarrow$ 1e-5	1e-5 $\rightarrow$ 1e-6
58.827%	11.905%	0.2180%

Any step size larger than 1e-5 is too coarse and will not give as accurate results as shown by the large percentage differences. Any smaller than 1e-5 does not give too much more accurate results, but the calculation costs more in terms of size and computing power. For the rest of the calculations, a step size of 1e-5 is used unless otherwise stated.

2.3 Radicals During Induction Phase of Reaction– Q2(c)

The radicals that are of importance in the induction phase are H, O, OH, and HO₂. As shown by Fig. 8, the magnitudes of the mass fraction in the induction phase are relatively constant at a

value of zero. The induction phase is the phase before the reaction, or in this case, before the spike in the mass fractions.

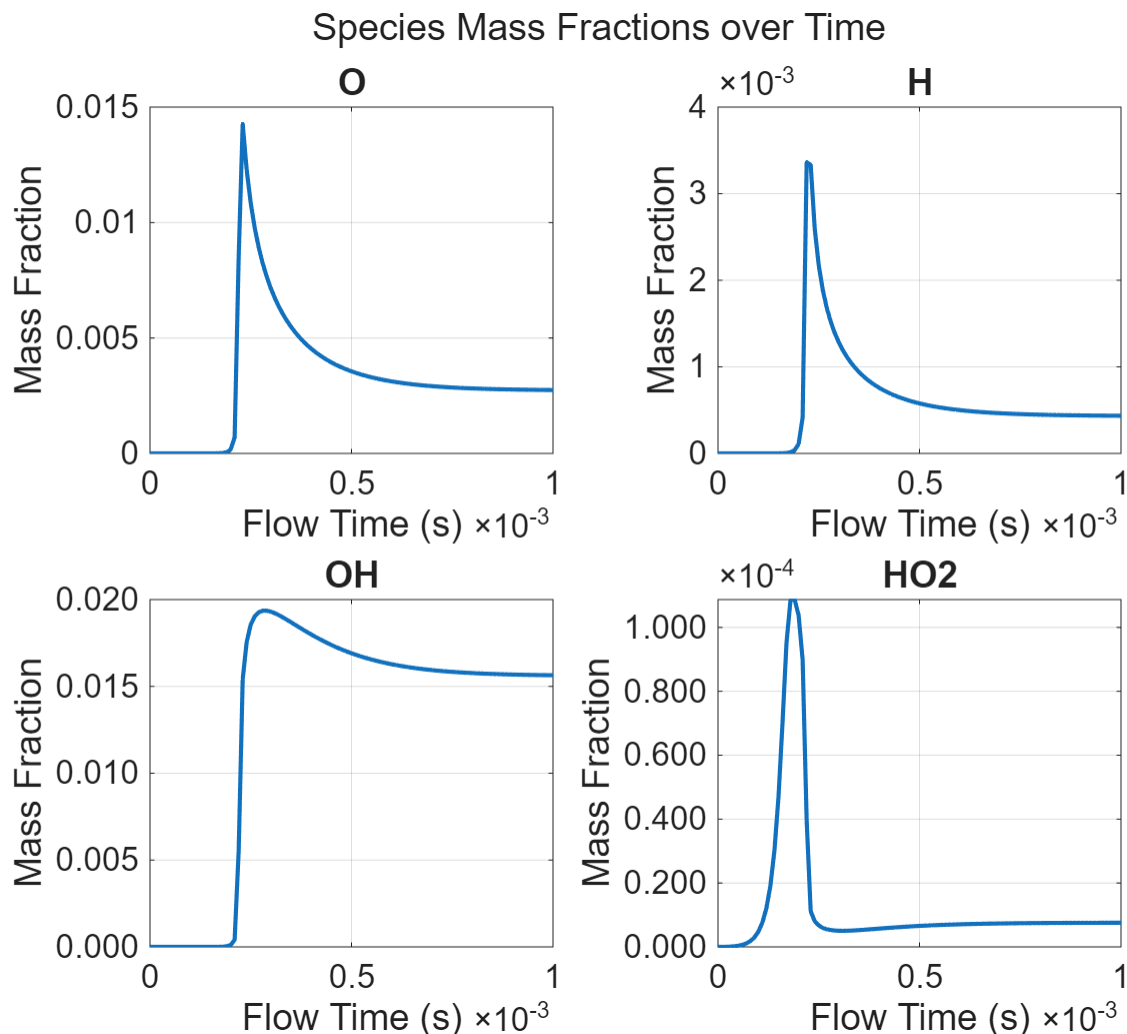


Figure 8 Radicals Mass Fractions

Out of all the radicals, it is clear from Fig. 8 that OH has the highest mass fraction after induction. The figure also shows that HO₂ quickly reacts, and it reacts almost in its entirety. This is not the case with OH where it has a higher mass fraction after the reaction meaning that it is a part of the products. Table 7 shows the time that it takes for each radical to meet 95% of its final value. This matches the timescale characteristics of the growth.

Table 7 Timescale Characteristics of Radicals (Time for Mass Fraction to Reach 95% of Max Value)

O	H	OH	HO ₂
2.300e-4	2.200e-4	2.500e-4	1.800e-4

It is evident in table 7 that the timescale for OH is the largest and for HO₂ it is the slowest meaning HO₂ reacts the fastest.

Fig. 9 shows the reaction rates with units $\text{kg}/(\text{m}^3 \cdot \text{s})$ for each reaction that includes HO_2 . The faster reactions contribute the most to the production of the species, and the higher the maximum reaction rate the higher contributor to the production of the products as well.

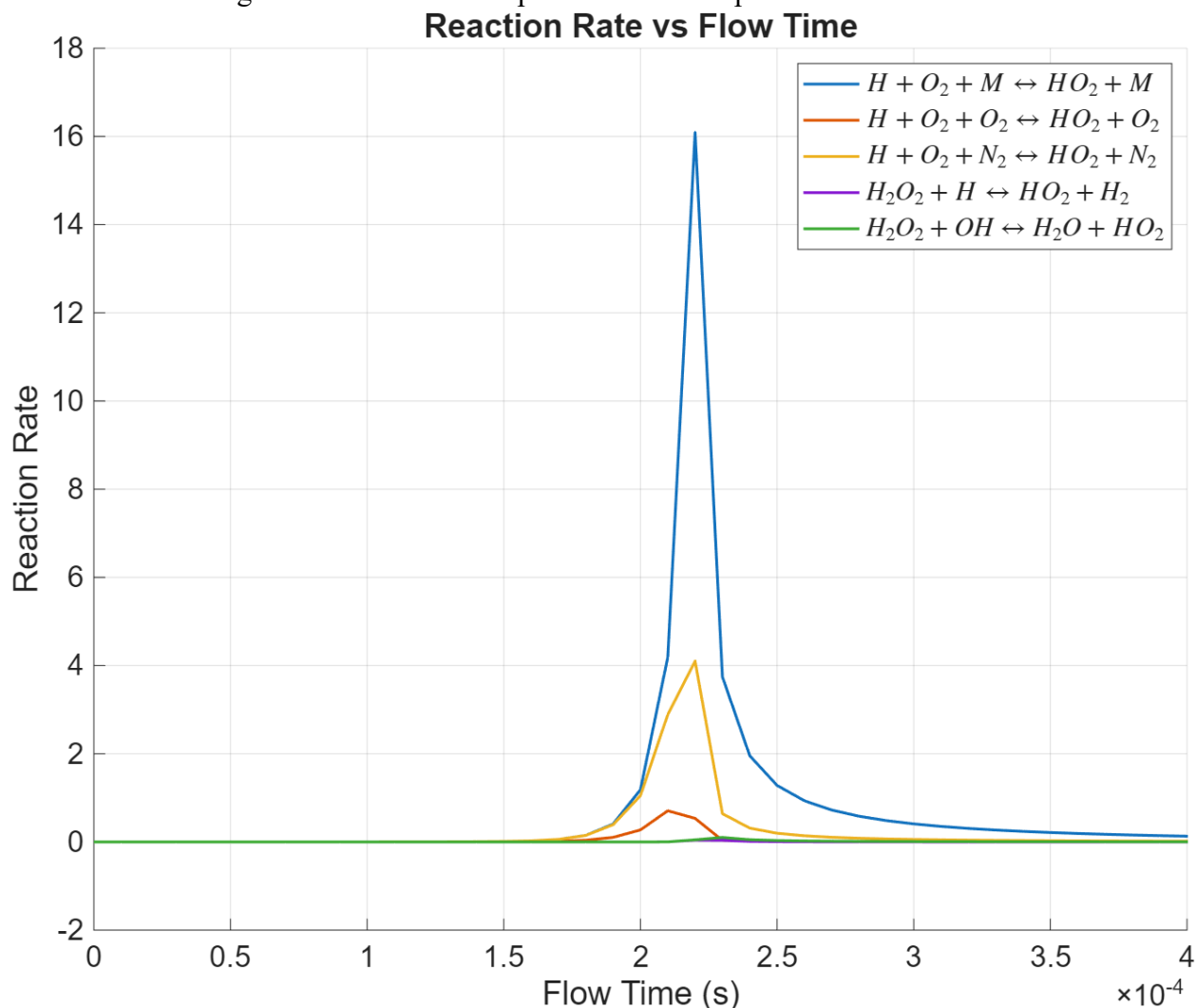


Figure 9 Reaction Rate vs Flow Time for HO_2

It is evident from Fig. 9 that the reaction of $\text{H} + \text{O}_2 + \text{M}$ contributes the most to the production of HO_2 . This is followed by $\text{H} + \text{O}_2 + \text{N}_2$, and the rest of the reactions are small enough that they are close to negligible.

Fig. 10 shows the reaction rates with units $\text{kg}/(\text{m}^3 \cdot \text{s})$ for each reaction that includes H . The faster reactions contribute the most to the production of the species, and the higher the maximum reaction rate the higher contributor to the production of the products as well.

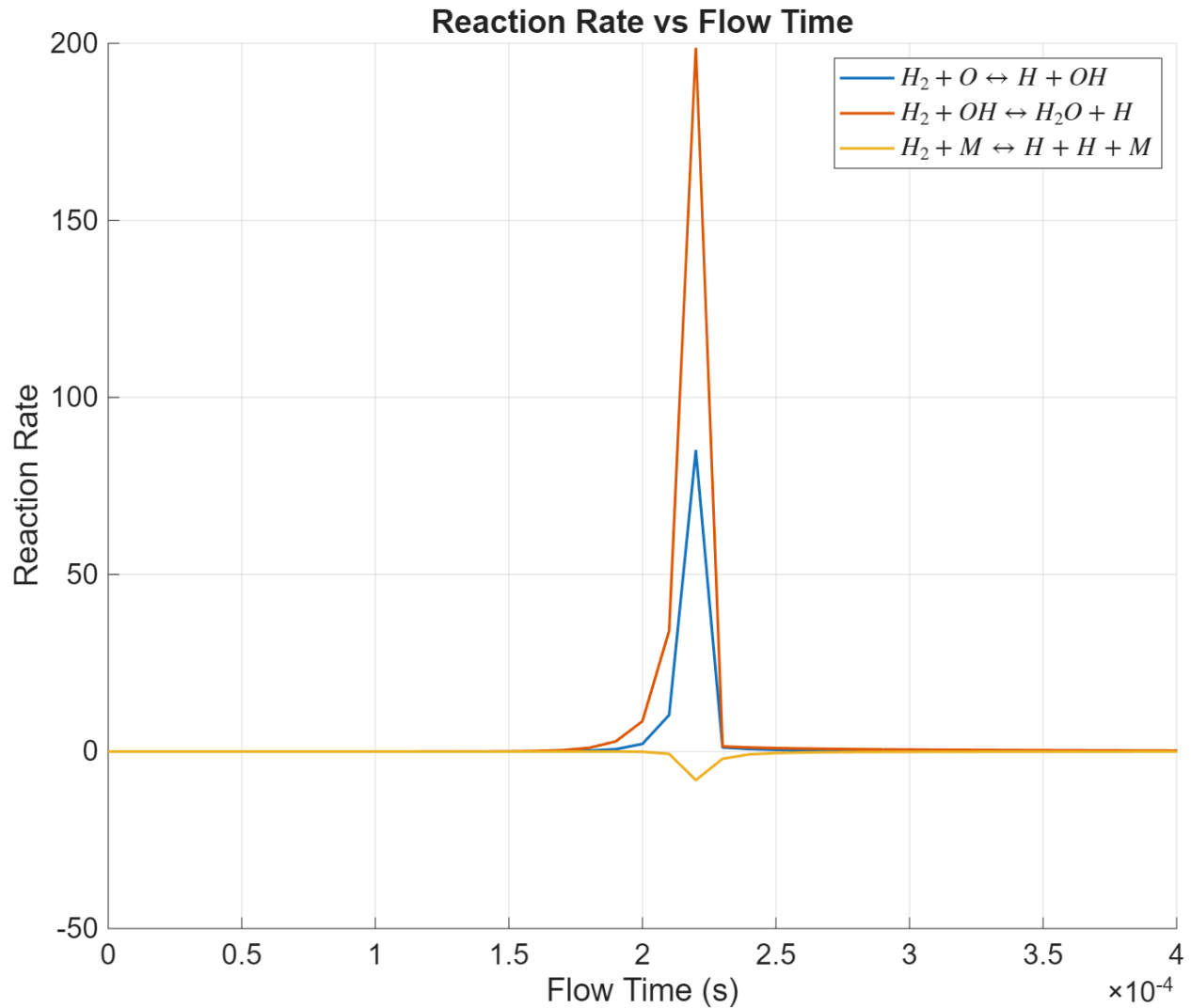


Figure 10 Reaction Rate vs Flow Time for H

It is evident from Fig. 10 that the reaction of $H_2 + OH$ contributes the most to the production of H. This is followed by $H_2 + O$, and the other reaction has a negative maximum reaction rate which means the species is being consumed rather than generated.

2.4 Fuel-Consumption Phane of Reaction – Q2(d)

Fig. 11 shows the mass fractions plotted for each species with the mass fraction of water to compare the consumption. It can be shown that N_2 is not affected by the creation of H_2O which makes sense because nitrogen is not in water.

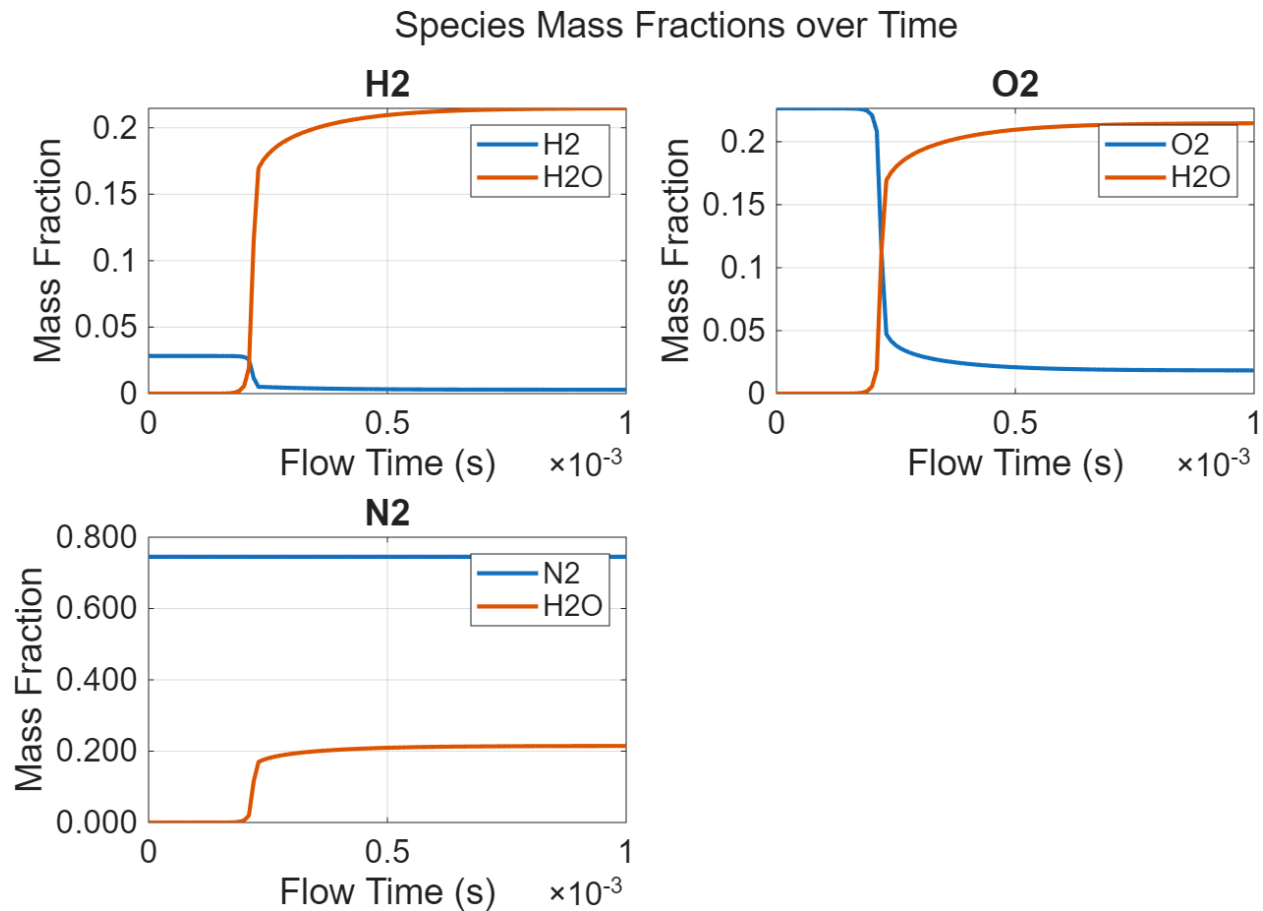


Figure 11 Comparison of Species Mass Fractions and Water Mass Fraction Over Time

The figure also shows that O₂ and H₂ are depleted while H₂O is created which makes sense. Table 8 shows the timescale associated with fuel consumption which is the time that it takes for the fuel to meet 95% of its final value. This matches the timescale characteristics of the growth.

Table 8 Timescale Characteristics of Fuel (Time for Mass Fraction to Reach 95% of Max Value)

H ₂	O ₂	N ₂
2.100e-4	2.100e-4	N/A

The timescale value for N₂ is not a number because the mass fraction of N₂ stays constant throughout the reaction. The timescales for H₂ and O₂ are the same which means they get depleted at the same rate.

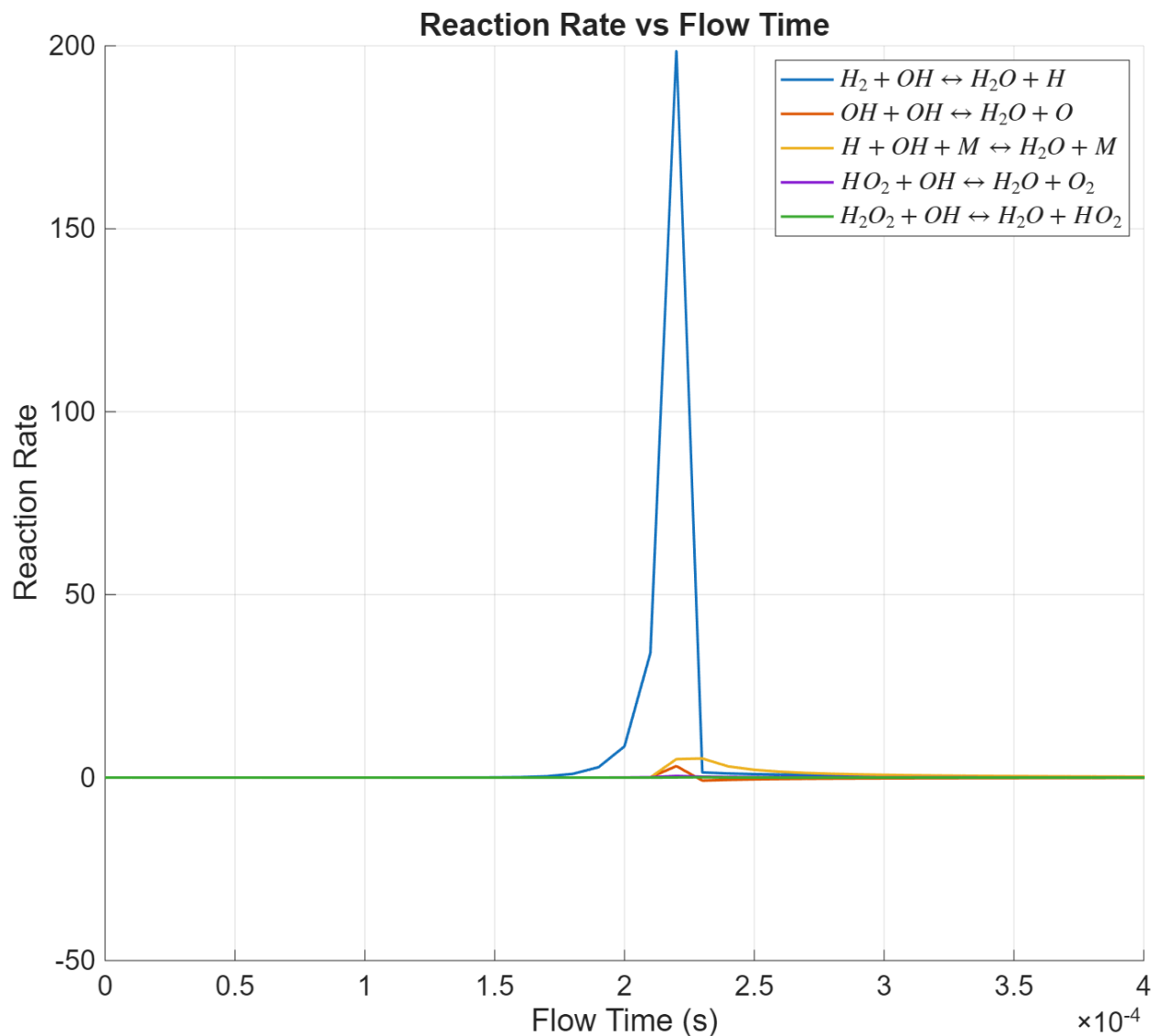


Figure 12 Reaction Rate vs Flow Time for H_2O

Fig. 12 shows the reaction rates with units $kg/(m^3 \cdot s)$ for each reaction that includes H_2O . The faster reactions contribute the most to the production of the species, and the higher the maximum reaction rate the higher contributor to the production of the products as well. It is evident from Fig. 12 that the reaction of $H_2 + OH$ contributes the most to the production of H_2O . The other reactions have negligible reaction rates. $OH + OH$ has a negative reaction rate being H_2O is being consumed rather than being produced.

2.5 Radical-Recombination Phase of Reaction – Q1(e)

The radicals that are of importance in the recombination phase are H , O , OH , and HO_2 . As shown by Fig. 13, the magnitudes of the mass fraction in the recombination phase are relatively constant at a value of zero. The recombination phase is the phase after the reaction, or in this case, after the spike in the mass fractions.

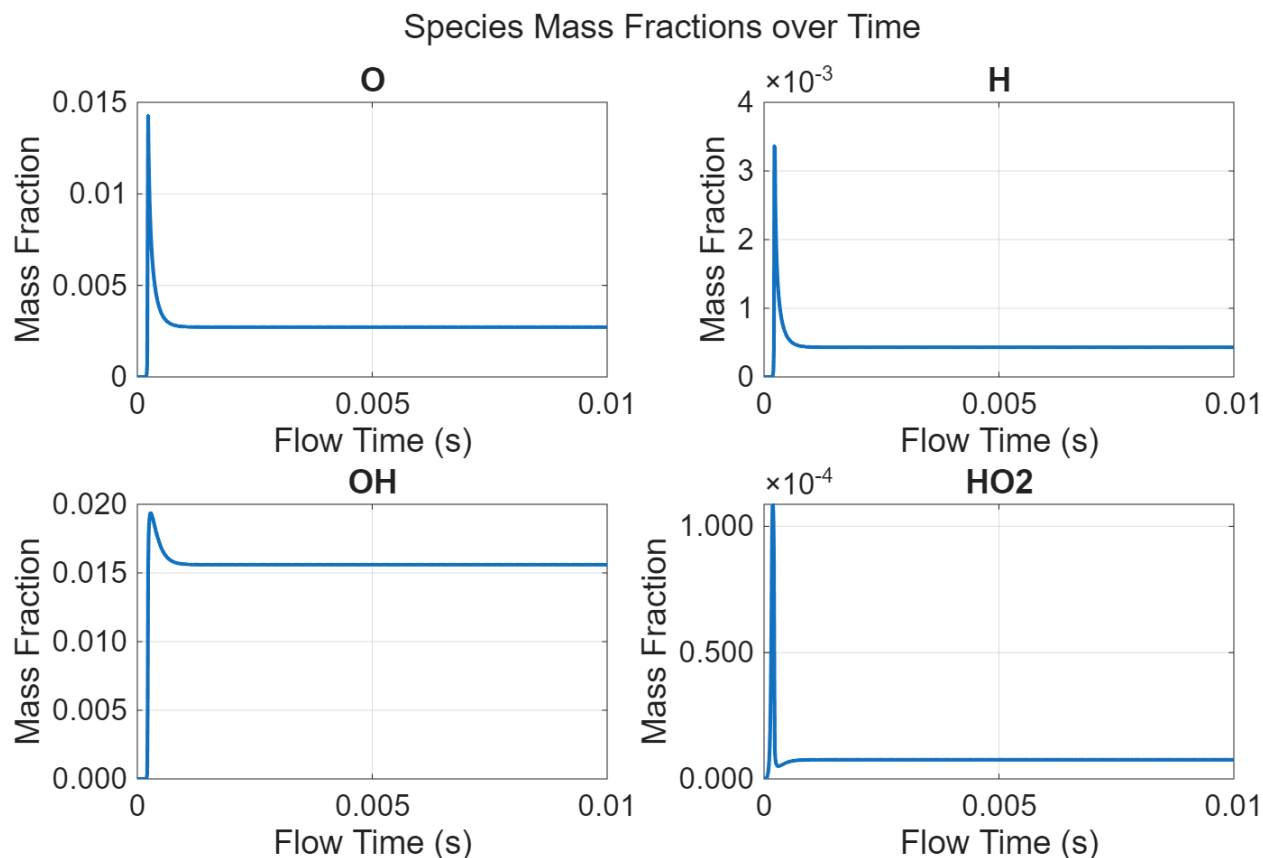


Figure 13 Radical Mass Fractions in Recombination

Out of all the radicals, it is clear from Fig. 13 that OH has the highest mass fraction in recombination and is thus the dominant radical. Fig. 14 shows the reaction rates with units $\text{kg}/(\text{m}^3 \cdot \text{s})$ for each reaction that includes H it is zoomed into an equilibrium position to better show the reaction rates in recombination. The higher the reaction rate the higher contributor to the production of the products as well.

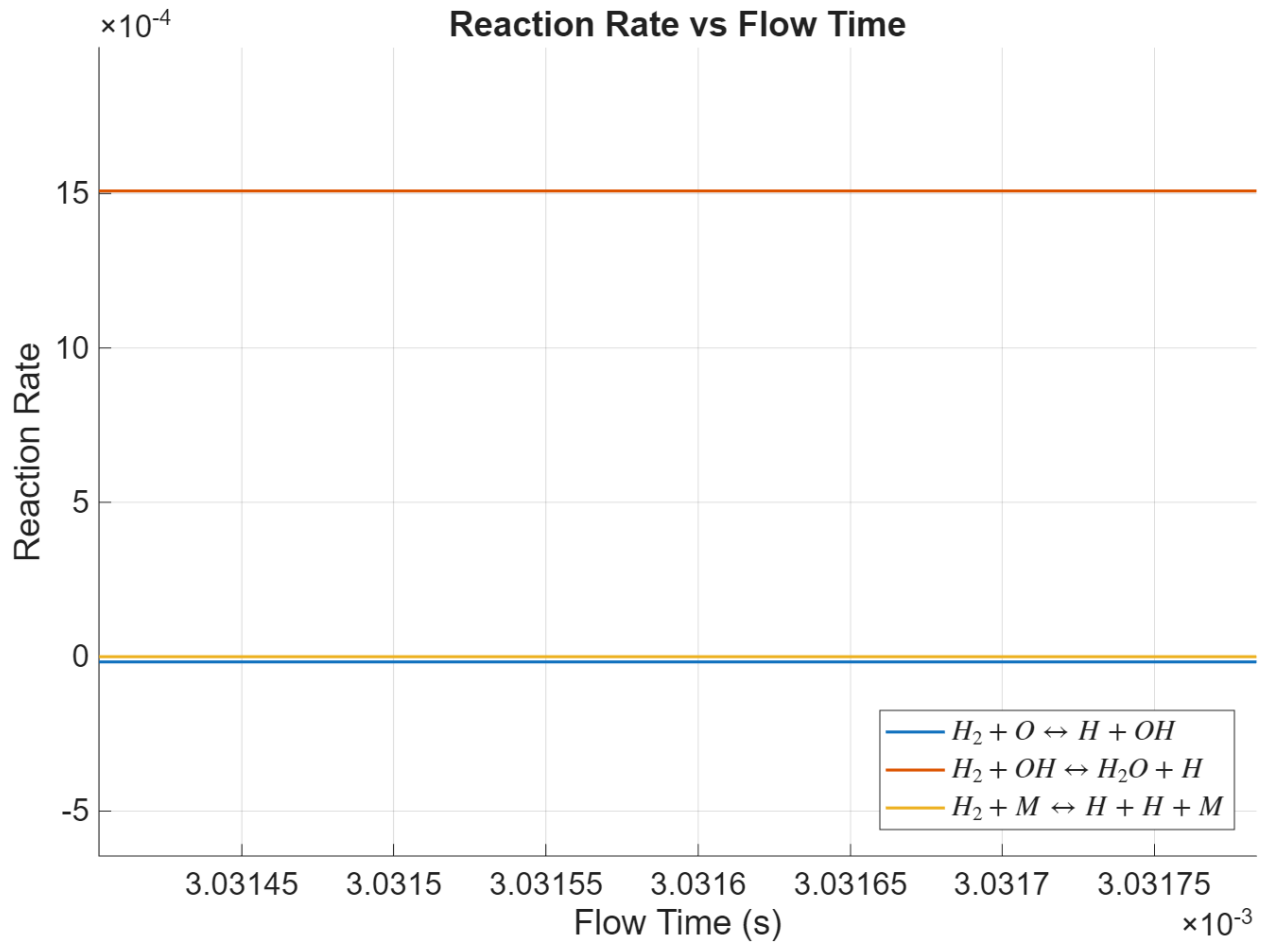


Figure 14 Reaction Rate vs Flow Time for H

Recombination occurs after the spike in reaction rate. It is evident from Fig. 14 that the reaction of $H_2 + OH$ contributes the most to the production of H as it has the greatest reaction rate after the spike. The other reaction rates are negligible as they are very close to zero. It is important to note that all the reaction rates are low in recombination, and Fig. 10 still is true for this scenario. Table 9 shows the time that it takes for each radical to lose 95% of its final value. This matches the timescale characteristics of the decay.

Table 9 Timescale Characteristics of Radicals (Time for Mass Fraction to Lose 95% of Max Value)

O	H	OH	HO ₂
N/A	0.0066	N/A	0.0099

It is evident in Table 9 that the timescale for HO₂ it is the largest meaning HO₂ reacts the fastest. The reason that O and OH do not appear in the table is that these radicals never loose 95% of their mass fractions. The time it takes for O and OH to drop from their maximum value to the equilibrium value is 0.00041 and 0.00032 seconds respectively.

2.6 Chemical Equilibrium Comparison - Q2(f)

If FLUENT runs for a long period of time, the system should be in equilibrium. Table 10 compares the equilibrium results of the mass fractions for the last point from FLUENT and the values from the Equilibrium_COSILAB.dat. The table also shows the percent difference of the results from FLUENT and COSILAB.

Table 10 Comparison of Mass Fraction Equilibrium Results

Solver	O	H	OH	HO ₂	H ₂ O	O ₂	H ₂	N ₂
FLUENT	2.7e-3	4.3133e-4	1.56e-2	7.5964e-6	2.148e-1	1.84e-2	2.9e-3	7.452e-1
COSILAB	2.6856e-3	4.4335e-4	1.5540e-2	7.3773e-6	2.1558e-1	1.7504e-2	3.0223e-3	7.4521e-1
Percent Difference (%)	0.5348	2.7484	0.13427	2.9265	0.36247	4.9911	4.1301	0.001342

The percent differences for each of the species are very small, the maximum percent differences being less than 5%. The species with the greatest percent differences are O₂ and H₂. This means that FLUENT does a good job of equilibrium conditions. This also means that there is a good chance that the other data that FLUENT reported is correct if it got the equilibrium conditions correct.